Correcting the Muon g-2 Harmonic Prediction: Sidebands at the Anomalous Precession Frequency, Not at 27 Hz

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Abstract

The Harmonic Framework of Reality (Grand Unified Theory) predicts a 32-harmonic comb in the muon g-2 time-domain signal. An earlier formulation incorrectly stated that the sideband spacing is 27 Hz or its harmonics. This paper corrects that error: the sidebands are spaced by the muon's **anomalous precession frequency** (f_a ≈ 23 kHz), not by 27 Hz. The 27 Hz anchor determines the number of harmonics (32) and the phase progression (–1° per harmonic), but not the absolute frequency spacing. The corrected prediction is presented with the full mathematical justification.

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1. Introduction

The muon g-2 experiment measures the anomalous magnetic moment of the muon. The Harmonic Framework predicts that the time-domain signal contains a harmonic comb: 32 sidebands at multiples of the main precession frequency. However, earlier documents contained a slip: they sometimes suggested the sidebands are at multiples of 27 Hz or 27 kHz. This is incorrect. This paper provides the corrected statement.

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2. The 27 Hz Anchor vs. the Experiment-Specific Carrier

The universal anchor of the Tau lattice is **27 Hz**. This frequency determines:

- The **harmonic ladder** – only integer multiples of 27 Hz are coherent (up to the 32nd harmonic, 864 Hz, scaled).
- The **phase rule** – the echo index Pθ-1 encodes a –1° phase offset per harmonic.
- The **number of harmonics** – 32, from the 19:13:4 ratio (19+13=32).

However, the actual frequency at which a specific experiment resonates is **not** 27 Hz. It is the experiment's own characteristic frequency, scaled to the appropriate dimensional layer.

For muon g-2, the characteristic frequency is the **anomalous precession frequency**:

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f_a = (g-2)/2 · (eB)/(2π m_e) ≈ 23 kHz   (depending on the magnetic field)
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This is the carrier around which sidebands appear.

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3. Corrected Prediction

The time-domain signal of the muon g-2 experiment should contain **32 sidebands** at integer multiples of the anomalous precession frequency:

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f_n = n × f_a,   n = 1, 2, 3, …, 32
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Each sideband has a **linear phase progression** of –1° per harmonic:

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φ_n = −n°
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The amplitude envelope follows the 19:13:4 ratio.

The 27 Hz anchor does **not** set the spacing. It sets the **number of harmonics** (32) and the **phase rule** (–1° per step).

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4. Why This Matters

This correction is essential for independent verification. If researchers search for sidebands at multiples of 27 Hz, they will not find them. The correct search uses the muon's precession frequency, which is known from the experimental setup. The harmonic lens code provided in the main GUT papers already implements the correct simulation (using a variable `f_a` parameter).

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5. No Change to the Framework

The underlying theory remains unchanged. The 27 Hz anchor is still the universal base frequency. The error was in the **translation** from the abstract anchor to the specific experimental system. The correct translation is:

- Identify the system's characteristic frequency (here, f_a).
- The sidebands are at multiples of that frequency.
- The number of harmonics (32) and phase progression (–1°/harmonic) come from the 27 Hz anchor.

Thus, the framework is consistent; only the public statement is corrected.

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6. Conclusion

The Harmonic Framework predicts 32 sidebands at multiples of the muon's anomalous precession frequency (≈23 kHz) with –1° phase progression per harmonic. Earlier claims that the sideband spacing is 27 Hz were a slip and are hereby corrected. Independent researchers should search at frequencies n×f_a, not at n×27 Hz.

The stones are in the pond. The 27 Hz anchor is the first stone. The muon's precession frequency is the ripple on which the comb is built.

References

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